

Global EV and Battery Manufacturing Capabilities

We do things right.
We challenge the accepted.
We aim higher.
We live inclusion.

Jacobs by the Numbers

At Jacobs, we're challenging today to reinvent tomorrow by solving the world's most critical problems for thriving cities, resilient environments, mission-critical outcomes, operational advancement, scientific discovery and cutting-edge manufacturing, turning abstract ideas into realities that transform the world for good.

With almost \$15 billion in annual revenue and a talent force of approximately 60,000, Jacobs provides a full spectrum of professional services including consulting, technical, scientific and project delivery for the government and private sector.

Company Facts

FOUNDED IN **1947**

60,000
TALENT FORCE

400
OFFICES

\$14.9B
ANNUAL
REVENUE

\$27.9B
BACKLOG

Net zero
CARBON FOR
OPERATIONS AND
BUSINESS TRAVEL

\$2.5B
CLIENT SAVINGS

Engineering News-Record 2023 Rankings

#1
TOP 500
DESIGN FIRMS

#1
MANUFACTURING

#1
INDUSTRIAL
PROCESS

#1
PROJECT MANAGEMENT/
CONSTRUCTION MANAGEMENT

#2
AUTO PLANTS

#2
ELECTRONIC ASSEMBLY

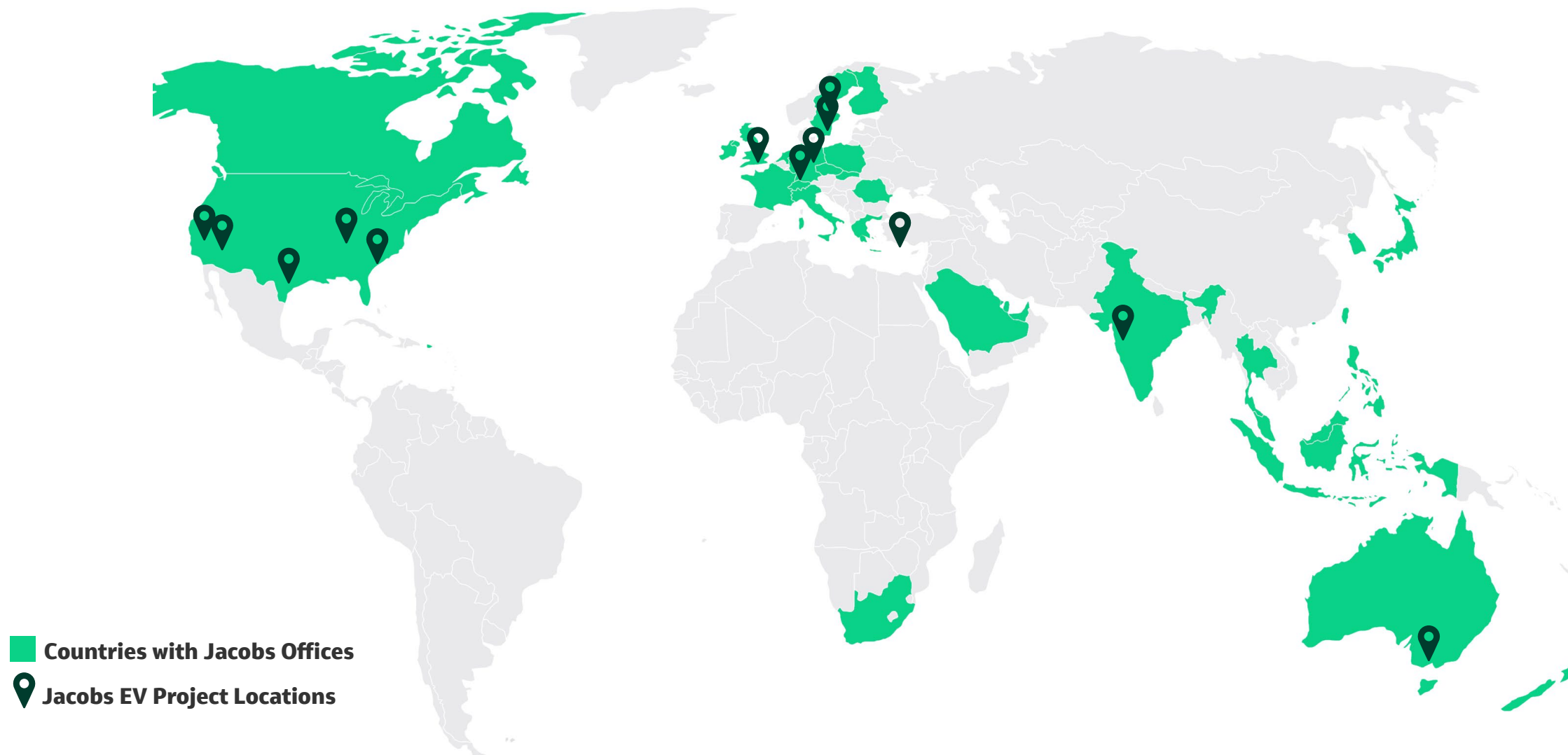
Battery Cell Manufacturing & Electric Vehicle Experience

- Supporting Global Electric Vehicle Manufacturers
 - Delivering three of the **largest Battery Cell Plants**, for a total of 350 GwH
 - Completed four of the **largest EV Manufacturing Plants** totaling 25 million square feet
- Electric Vehicle Manufacturing Optimization Expertise

***Working together with our client,
we delivered an EV manufacturing
and battery plant in North America
in 18 months***



Supporting EV Projects Globally



60,000+ employees | 40+ countries | 400 offices

Capabilities – Battery Cells & EV Manufacturing

Facilities Design & Build

- Site Selection and Master Planning
- Environmental & Permitting Support
- Architectural
- MEP
- Civil
- Structural
- Design for Manufacturing
- Concurrent Process and Facilities Design

Mechanical Process

- High pressure and high temperature liquid and gas systems
- Refrigeration systems
- Hypergolic and cryogenic fuel and oxidizer systems

Machine Design

- Hydraulic systems
- Custom machine design
- High speed rotating equipment
- Load measurement systems
- Mechanical positioning & handling systems
- Static/dynamic/thermal FEA

Electrical

- Power engineering analysis and design
- Grounding

IT and Analytical Services

- Network design and engineering
- Cyber security
- Predictive and analytical modeling
- Operational decision support tools

Instrumentation

- Data acquisition
- Signal conditioning

Control Systems

- Distributed PLC systems
- ISA instrumentation specifications

Software

- Custom application development
- Commercial off-the-shelf integration
- Web-based and real time applications
- Networking

Construction and Integration Services

- Construction management
- Procurement
- Hardware build-up
- Pre-installation integration/testing
- Software development
- Field start-up and commissioning

Manufacturing Engineering

- Concurrent product and process design
- Lean manufacturing/optimization
- Industrial engineering
- Launch and operations support
- Material and logistics support

Operations

- Enterprise O&M
- Technical Support

Robotics

- Process design and optimization
- Tooling design
- Robot programming
- Training

Maintenance

- Process assessment
- Preventive maintenance
- Predictive technology
- Reliability analyses

Metrology

- Measurement uncertainty analyses

Safety

- System safety design/analyses
- Occupational/construction safety
- Facility siting

Structures

- ASME pressure vessels
- ANSI B31 piping
- Flexibility and finite element analysis
- Structural dynamics
- Hardened structures
- Special movable structures
- Concrete, steel, composites, and specialty alloys

Battery Project Experience

Client	Project	Location
Confidential Client	Gigafactory for EV Car and Cell Manufacturing	Texas, USA
Confidential Client	Gigafactory for EV Car and Cell Manufacturing	Germany
Confidential Client	Lithium Ion Battery Manufacturing	Nevada, USA
Energizer Power Systems	Lithium Ion Battery Manufacturing	Gainesville, Florida, USA
Boston Power	BOD, cost estimate, site selection, environmental	Westborough, Massachusetts, USA
Molecular Nanosystems	US DOE Battery Development Center Grant Application Support	Paramus, New Jersey, USA
Infinite Power Solutions	Thin Film Battery Manufacturing	Littleton, Colorado, USA
EnerG2	Nano-engineered high-performance carbon electrode material	Albany, Oregon, USA
Duracell	Material Tracking Computer System	Lancaster, South Carolina, USA
Duracell	Cell Aging and Supply Chain Storage	Lancaster, South Carolina, USA
Matsushita Battery	Waste Management Plan	Columbus, Georgia, USA
Matsushita Battery	Packaging and Finished Goods Warehouse	Columbus, Georgia, USA
General Electric	Nickel and Cadmium Reclamation	Gainesville, Florida, USA
Panasonic	Nickel-Cadmium Battery Recycling Study	Secaucus, New Jersey, USA
Saft	Experimental/Pilot Battery Plate Manufacturing	Valdosta, Georgia, USA
Seitzinger	Recycling facility	Atlanta, Georgia, USA
National Alliance for Advanced Transportation Batteries	US DOE Development Center Grant Application	Chicago, Illinois, USA
Xtreme Power	Modification of Ford Motors manufacturing plant	Wixom, Michigan, USA

Automotive Experience

Our automotive and advanced manufacturing team has been working with automotive manufacturers as well as tier one industry suppliers to meet their business objectives for decades.

Our highly specialized skills and valuable experience enable us to provide integrated solutions with vast resources.

TIER ONE SUPPLIERS

American Axle and Manufacturing	Driveline Manufacturing Plant	Jiangsu Province, China	376,600 sf.
Denso	Instrumentation Cluster and Electronic Plants	Marysville, TN	30,000 sf.
GKN Automotive Inc.	Drive Line Manufacturing Plant	Auburn Hills, MI	265,000 sf.
Hino Motors (Obayashi)	Axle Plant	Marion, AR	1,600,000 sf.
John Deere	New Engine Plant	Torreon, Mexico	538,000 sf.
PBR	Brake Components Greenfield Plant	Knoxville, TN	218,000 sf.
Plastic Omnium	Thermal Injection Molding Facility	Anderson, SC	100, 000 sf.
Rockwell Automation	Aluminum Wheel and Truck Facility	Laurinburg, NC	223,000 sf.
TRW	Automotive Brakes System Plant	Queretaro, Mexico	215,000 sf.
TRW-Koyo Steering	Bulk Storage Facility	Vonore, TN	550,000 sf.

AUTOMOTIVE

BMW	North American Manufacturing Complex	Spartanburg: SC	215, 000 sf.
BMW Brilliance	Assembly Plant	Shenyang, China	753,000 sf.
Confidential Client	Transmission Factory	Mexico	300,000 sf.
Daimler Chrysler	Manufacturing Plant	Mexico City, Mexico	1,500,000 sf.
Daimler Chrysler	Vehicle Assembly Plant	Savannah, GA	1,500,000 sf.
General Motors	Office Building Complex	Mexico City, Mexico	150,000 sf.
Honda	Automobile Manufacturing Plant	Lincoln, Alabama	4,300,000 sf.
KIA Motors	Automotive Assembly Plant	West Point, GA	1,500,000 sf.
Peugeot Citr�en	Automotive Plant	Wuhan, China	400,000 sf.
Nissan Motor Company	Automotive Plant	Aguascalientes, Mexico	550,000 sf.
Nissan Motor Company	Power Train Assembly Plant	Decherd, TN	1,500,000 sf.
Toyota	Ann Harbor Technology Park	Ann Harbor, MI	750,000 sf.
Toyota Motor Sales	Automotive Parts Distribution Center	Hebron, KY	792,000 sf.

HINO Motors
Marion, Arkansas, USA
Automotive Axle Manufacturing



Confidential Client
Reno, Nevada, USA
Lithium Ion Manufacturing



Confidential Client
Mexico
Automobile Manufacturing



KIA Motors
West Point, Georgia, USA
Automotive Assembly



BMW
San Luis Potosi
Automotive Assembly



Honda
Lincoln, Alabama
Automotive Assembly



Relevant Project Examples



New Electric Vehicle and Battery Facility



NEW EV AND BATTERY MANUFACTURING FACILITY INCLUDING:

- Stamping, Casting, Body in White, Paint, and General Assembly
- Cell production and Battery Assembly
- Drive Unit Manufacturing

SIZE:

- Total Building Area 11,500,000 ft²
- Cell Building Area 1,000,000 ft²
- 100 Gwh with Phase 2

SERVICES: Design and Field Support Services

PROJECT DURATION: Ongoing

LOCATION: North America

Project Description

- Gigafactory for EV Car and Cell Manufacturing
- EV Car manufacturing from raw material to stamping and casting, paint, and to final assembly.
- Complete cell manufacturing process from raw material delivery and mixing, coating and drying, electrode manufacturing, through assembly and formation
- 100,000 ft² of employee office, break room, conference room, rest room, and dining spaces.
- Battery pack and drive unit assembly
- Plastics Injection

Services Provided by Jacobs

- Architectural, Civil Land Development, Waster Water, Industrial Water, Potable and Process Water, Transportation, Structural, Mechanical, Electrical, Process and Controls Engineering
- Building information modelling
- Design of specialist systems including electrolyte distribution, dust collection, VOC abatement, clean room and dry air systems.
- Smoke and fire modeling for life safety
- Permitting support
- Field engineering and construction management support

Key Drivers

- Speed to Market, Flexibility, Nimble Response, Specialized Resources, Ability to Right Size Team

New Photovoltaic Cell Production and Module Assembly



NEW CELL MANUFACTURING AND PHOTOVOLTAIC MODULE ASSEMBLY:

- Cell Production
- PV Module Assembly
- Office/Administrative and Support Buildings

SIZE:

- Phase 1 – 3GW with two 1.5GW parts
- Phase 2 – expansion to 6GW

SERVICES: Design and Field Support Services

PROJECT DURATION: 2022 – ongoing

LOCATION: USA

Project Description

- Site due diligence, master planning, permitting and design services for a 6GW cell production and PV module assembly Gigafactory in the United States
- New campus is planned on a 400-acre greenfield site, with approximately 4 million square feet of production, administration and support buildings
- The project is phased, with a capacity of 3GW by 2024 and a full buildout capacity of 6GW by 2025
- Production is expected to be on a 24/7/365 schedule

Services Provided by Jacobs

- Site Due Diligence (multiple potential locations assessed)
- Site-Agnostic Basic Design
- Site-Specific Master Planning and localization of Basic Design
- Permitting support
- Support studies (Water/Wastewater, Environmental Assessment for DOE LPO)
- Designer for Design-Build, construction and commissioning support

Key Drivers

- Highly collaborative approach with owner, contractors, trade partners and vendors, aggressive schedules, delivery to a target budget, team agility, site adaptation and localization of non-US design

New EV Cell and Battery Facility



NEW EV CELL AND BATTERY MANUFACTURING FACILITY INCLUDING:

- Cell production
- Battery Assembly
- Drive Unit Manufacturing

SIZE:

- Total Building Area 1,000,000 ft²
- 100 Gwh with Phase 2

SERVICES: Design and Field Support Services

PROJECT DURATION: Ongoing

LOCATION: Europe

Project Description

- Gigafactory for EV Cell and Battery Manufacturing
- Complete cell manufacturing process from raw material delivery and mixing, coating and drying, electrode manufacturing, through assembly and formation
- Quality control laboratories
- Battery pack and drive unit assembly
- Offices, social spaces and support equipment

Services Provided by Jacobs

- Mechanical, Electrical, Process and Controls engineering
- Building information modelling
- Design of specialist systems including Industrial heating and cooling plant, electrolyte distribution, dust collection, clean room and dry air systems.
- Permitting support
- Field engineering and construction management support

Key Drivers

- Speed to Market, Flexibility, Nimble Response, Specialized Resources, Ability to Right Size Team

Electric Vehicle Cell and Battery Facility



EXPANSION OF A LARGE-SCALE LITHIUM-ION BATTERY FACTORY DESIGNED TO MANUFACTURE BATTERIES FOR ELECTRIC VEHICLES AND ENERGY STORAGE SOLUTIONS:

- Cell production
- Battery Assembly

SIZE:

- Total Building Area: 1,500,000 sqm
- 61 Gwh

SERVICES: Integrated Design Services including Basic and Detailed Design

PROJECT DURATION: Ongoing

LOCATION: Europe

Description of the project scope

- Large-scale EV battery manufacturing facility
- Full site expansion design for all buildings, including all ancillary buildings such as energy recovery, can stamping and warehousing
- Clean and dry rooms, utilities (spaces for compressed air and nitrogen handling), piping and relevant instrumentation, wastewater pre-treatment and storage areas, fire safety systems, HVAC, electric energy distribution and generation systems, an IT network and its relevant interfaces

Services provided by Jacobs

- EPCm Cost Estimation
- Design Coordination
- Risk assessments
- Constructability reviews
- Constructability and Maintenance Optimisation
- Providing Clean and Dry room contracting options
- Supporting cost optimisation sprints

Key Drivers

- Speed to Market
- Flexibility
- Specialized Resources
- Ability to Right-Size Team

Vehicle Assembly Plant

PROJECT COMPONENTS:

- Throughput Performance Assessment
- Maintenance Effectiveness Evaluation
- Recommendation Implementation

SERVICES: Throughput Optimization

PROJECT DURATION: Complete

LOCATION: Spartanburg, SC

Description of the project scope

- Led a cycle time and throughput improvement project for BMW's Spartanburg, SC facility
- Improved throughput in the underbody systems for the F15 platform, specifically for the engine compartment, firewall, wheelhouse, and floor pan lines
- Efficiently and accurately analyzed the systems, developed key deliverables, and implemented required actions

Services provided by Jacobs

- Completed thorough assessment of the overall manufacturing process and health of the equipment to determine if/where additional equipment would be required
- Identified areas of preventative maintenance to increase overall OEE of the underbody system
- Completed a thorough analysis of the robot equipment and processes to identify opportunities for processing, movement, and interference zone optimization
- Implemented the required actions which included: robot path improvement and optimization; robot load data correction; weld spot sequencing; robot I-zone optimization; mechanical modification of robots and tooling; correction of process inefficiencies and addition of robots in key areas; improved robot configuration and dress issues

Existing Electric Vehicle and Battery Facility



EXISTING EV VEHICLE AND BATTERY MANUFACTURING FACILITY:

SIZE:

- Total Assembly Building Area 3,600,000 sq ft
- Site Area: 403 acres

SERVICES: Project Management, Design and Field Support Services

PROJECT DURATION: Ongoing

LOCATION: North America

Project Description

We developed the design for miscellaneous retrofits and new equipment installation and continues to provide field support services for this existing EV and Battery Manufacturing Facility. The project scope includes:

- Lead Legacy Permit Closeout effort. 700+ open permits closed out in two years.
- Developed Assembly Building, South Paint, and Plastics buildings overall egress permit documents. All have been approved by the City
- Miscellaneous improvements related to manufacturing tools, workplace, restrooms, office, and cafeteria spaces
- Build working relationship with city building officials
- Hazmat Designs - Performance Based Designs
- Parking Improvement

Services Provided by Jacobs

- Design and Field Support Services including Architectural, Civil, Fire, Structural, Electrical, Mechanical, Plumbing, Process Piping, Landscape, BIM and Design Manager services.
- Smoke, egress, and fire modeling for life safety.
- Provide recommendations for hazardous materials quantities.
- Permitting support.
- 3D REVIT modeling and laser scanning.
- Supplement client plant design architectural/engineering team with our staff.
- Project performance reports to client leadership.

Key Drivers

- Improve relationship with City building officials, increase efficiency, improve quality, reduce review cycles with city plan reviewers, schedule, budget

Electric Battery & Drive Unit Facility



NEW BATTERY MANUFACTURING FACILITY INCLUDING:

- Cell Production and Battery Assembly
- Drive Unit Manufacturing

SIZE:

- Total Building Area: 4,900,000 ft²
- 350 Gwh

SERVICES: Design and Field Support Services

PROJECT DURATION: Ongoing

LOCATION: North America

Project Description

- Battery Cell and Drive Unit Manufacturing
- Complete cell manufacturing process from raw material delivery and mixing, coating and drying, electrode manufacturing, through assembly and formation

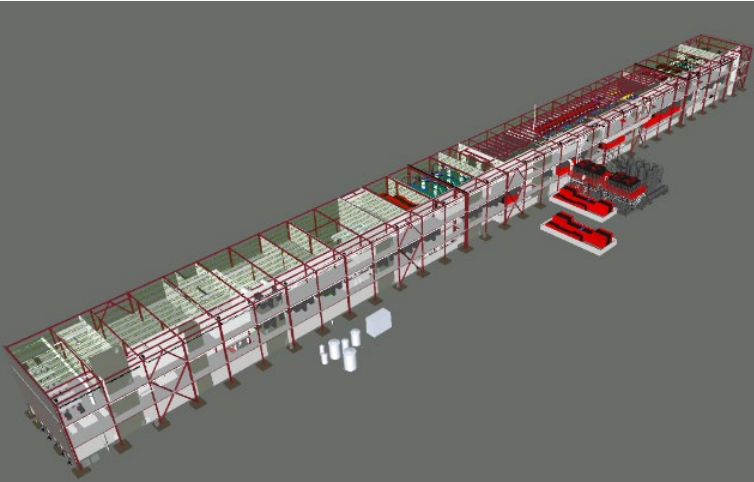
Services Provided by Jacobs

- Architectural, Civil, Waster Water, Industrial Water, Potable and Process Water, Transportation, Structural, Mechanical, Electrical, Process and Controls Engineering
- Building information modelling
- Design of specialist systems including electrolyte distribution, dust collection, VOC abatement, clean room and dry air systems.
- Smoke and fire modeling for life safety
- Permitting support
- Field engineering and construction management support

Key Drivers

- Speed to Market, Flexibility, Nimble Response, Specialized Resources, Ability to Right Size Team

EV Cathode Process Equipment



NEW EV BATTERY CATHODE MANUFACTURING FACILITY INCLUDING:

- Precursor production areas
- Bulk Powder Handling, including Continuous Mixing,
- Calcination, Brick Crushing, Washing/Filtration and QC
- Material Loading

SIZE: Cathode Building Area: 120,000 sq ft

SERVICES: Design Services

PROJECT DURATION: Ongoing

LOCATION: North America

Project Description

- Cathode Process Equipment Design
- Complete cell manufacturing process from raw material delivery and mixing, coating and drying, electrode manufacturing, through assembly and formation

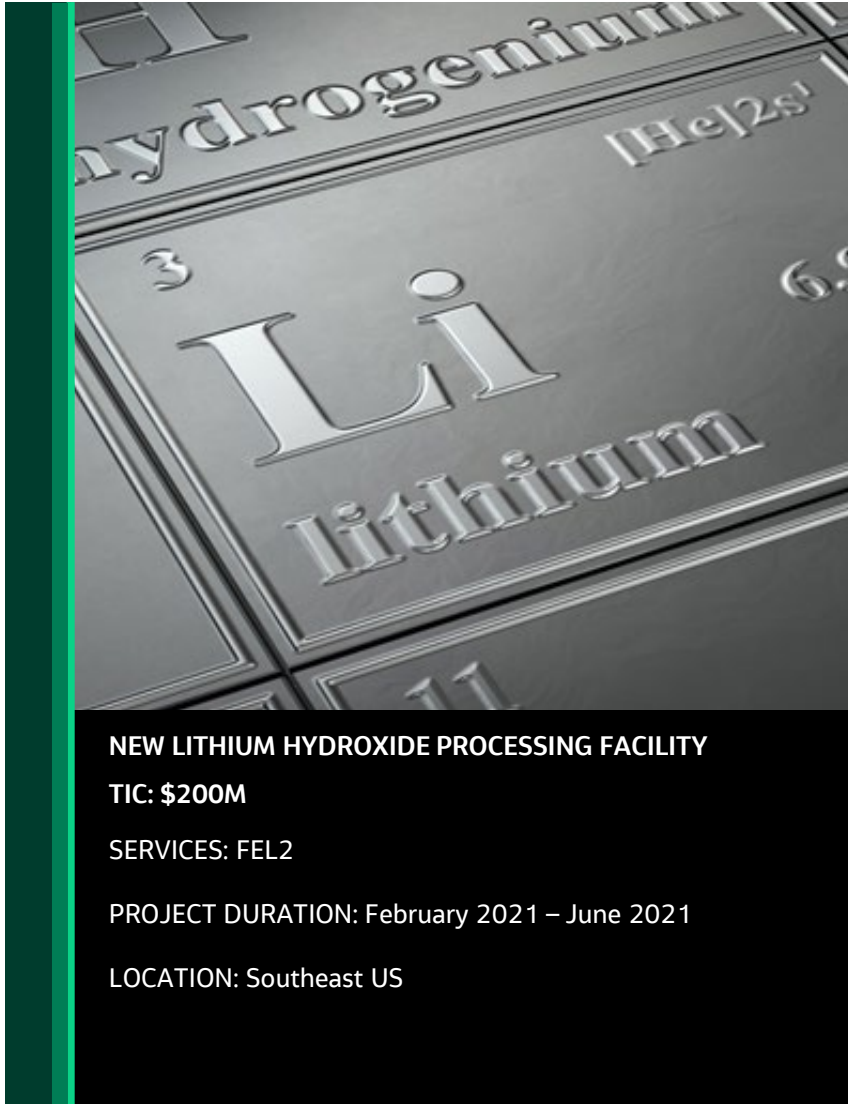
Services Provided by Jacobs

- Mechanical, Process, Piping and Controls engineering services
- Subject Matter Expertise (SME) services
- Process & Instrumentation (P&ID) Diagrams, Process Flow Diagram (PFD)
- 3D modeling using Revit, Plant 3D, SolidWorks and BIM360 collaboration
- Equipment specifications and datasheets development
- Controls functional description development

Key Drivers

- Scaling up a prototype plant to full production deploying a new technology

EV Battery Process - Lithium Hydroxide



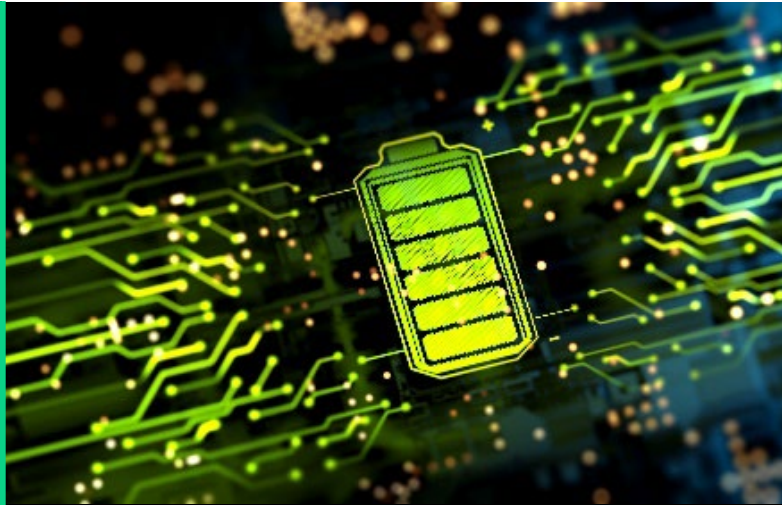
Project Scope

- Develop 20,000 tpa lithium hydroxide production facility conceptual engineering with capital cost and operating cost estimates and construction schedule
- Produce high-quality lithium hydroxide monohydrate ($\text{LiOH} \cdot \text{H}_2\text{O}$) powder, converted from varying impurity grade lithium carbonate material
- Project processes include the following:
 - Material Handling / Unloading
 - Mixing/reaction
 - Evaporation
 - Crystallization
 - Centrifugation
 - Drying
 - Packaging
 - Wastewater Treatment
- Greenfield option including production system & facilities, administration, laboratory, maintenance, utilities, raw & finish product warehousing and security
- Brownfield option at an existing mine processing site
- Use of our Global Integrated Design (GID) Team for an estimated client savings of \$20,000

Services Provided by Jacobs

- Conceptual Design
- Construction Cost Estimate for both Greenfield and Brownfield options
 - Aspen Plus Modeling
 - Process Flow Diagrams
 - Equipment Lists
 - Conceptual Layouts
 - Budgetary Equipment Pricing
 - Preliminary Electrical Load Requirements

Li-Ion Battery R&D Facility Cell Pilot Line



LI-ION BATTERY R&D FACILITY INCLUDING:

- Large Tooling serving Pilot Line of lithium-ion battery manufacturing process
- Pilot Scale Production
- Research & Development for targeted improvements and next generation tool development

SIZE: Building 1 (232,000 SF) and Building 2 (121,000 SF)

SERVICES: Detailed Design Services

PROJECT DURATION: Ongoing

LOCATION: Northwestern US

Project Description

We coordinated with the client's team of designers and engineers to provide detailed design services for Phase 1 tenant improvement of the Li-Ion Battery R&D Facility Cell Pilot Line. The cell pilot programs consist of lithium-ion battery pilot manufacturing lines referred to as:

- Large Tooling serving Pilot Line of lithium-ion battery manufacturing process
- Pilot Scale Production
- Research & Development for targeted improvements and next generation tool development

Services Provided by Jacobs

- Architectural and engineering services to support tenant fit-out and associated exterior and site improvements

Key Drivers

- Cost, Speed to Market

LiPF6 Sample Plant



HONEYWELL LiPF6 SAMPLE PLANT:

SIZE: 30 GWh / year

SERVICES: Basic Design, Procurement, Construction Management

LOCATION: Buffalo, NY

Project Description

We provided design for an installation of a new Lithium Salt product for the Lithium-Ion Battery Market, in an existing pilot plant. Project overall total installed cost was \$15 million.

Our scope included renovating, modifying, and reusing four floors of a multi-story building for processing phosphorous, hydrogen fluoride, fluorine, lithium and intermediate products.

The pyrophoric nature of phosphorous required jacketed piping and the PF5 transport required a specialized compressor.

The process included:

- One gas phase reactor step
- One counter current column reactor step
- One distillation step with final product isolated and dried on a nutsche filter dryer that produces a 70 kg/batch of lithium-ion battery electrolyte salts

Special metallurgy was required due to both corrosivity and high and low temperatures in portions of the plant. Temperatures less than -40oC are employed for crystallization using a specialized refrigeration system

Detailed design packages for all disciplines were provided to solicit lump sum pricing from contractors for the installation. 3D design was utilized to produce construction packages. Construction management was provided for all installation contracts.

Services Provided by Jacobs

- Basic design (FEL3)
- Procurement
- Construction management

Mercury Cell Conversion Project



MERCURY CELL CONVERSION PROJECT:

CLIENT NAME: ASHTA Chemicals

OUR ROLE: EPCM

CONTRACT TYPE: Time & Materials

CONSTRUCTION/TIC COST: \$120 million

PROJECT DURATION: 2017 to 2020

SIZE: 75,000 sf of inside and outside process area

OUR EXECUTING OFFICE/S: Greenville, SC and Mumbai, India (GID)

Project Description / Services Provided

The ASHTA Chemicals Plant, located in Ashtabula, Ohio, is roughly one mile from the shores of Lake Erie. ASHTA is a major producer of potassium hydroxide, potassium carbonate, hydrochloric acid, potassium hypochlorite, chlorine, and chloropicrin in North America. We provided conceptual, preliminary, and detailed engineering, procurement, and construction management services. We engaged our Mumbai, India Global Integrated Delivery (GID) office to execute approximately 50% during the design phase.

This project relied on multiple process systems working in a continuous process to produce intermediate products that were incorporated into the finished product. The process equipment was a mixture of technology packages, skid-mounted units, and stand-alone equipment fully integrated into the overall plant design. The process also required temperature and pressure control, careful materials of construction evaluation, and considerations during the detailed design phase that included line sloping and coordinating piping and equipment elevations.

The project converts the existing mercury cell line to BICHLOR membrane cells. The design basis capacity for the new membrane cells are based on two electrolyzers, each operating at 4.61 kA/m² (start of life, SOL). On this basis, the plant achieves design capacity of 60,000 STPY chlorine at 350 operating days per year. This equates to 274 STPD of KOH on a 100% KOH basis.

This new retrofit addition to the existing chlor-alkali facility has a minimum of 20-year life expectancy basis, with respect the materials of construction selection and corrosion allowances. The project required numerous tie-ins to be put in place during a pre-scheduled shut-down.

Most of the new process units were constructed in an area to the south of the existing plant. This required new infrastructure to support the new process. The new infrastructure consists of a pre-engineered process building, pre-engineered boiler building, some underground utilities, roads, pipe racks, standby generator, power and primary feeders.

Lithium Carbonate Plant Expansion



Project Description / Services Provided

We worked with Rockwood Lithium on a Lithium Carbonate Plant Expansion. The plant produced lithium carbonate via a precipitation reaction involving purified lithium chloride brine and a sodium carbonate (soda ash) solution. Subsequently, a production operation to produce lithium chloride anhydrous crystals and an additional operation to produce high-purity lithium carbonate via re-crystallization process were added. We provided conceptual, preliminary, and detailed engineering, procurement, and construction management services.

- The project required multiple unit operations with different types of product processing to form the final product. The project also required multi-continent project execution, which led us to work with our office in Santiago to help bridge the gap with local engineering and construction management resources.
- Our team achieved zero recordables and zero first aids during the project, winning the Santiago office a BeyondZero safety award.
- Through the design and construction of the project, we were able to deliver \$13.4 million in cost saving to Rockwood.

Total Installed Cost (TIC): \$200 million

Battery Recycling Program



BATTERY RECYCLING PROGRAM:

CLIENT NAME: Cirba Solutions

OUR ROLE: EPCM

CONTRACT TYPE: Time & Materials

CONSTRUCTION/TIC COST:

Project 1 Lancaster: \$200 million brownfield expansion

Project 2 Eloy: \$80 million brownfield retrofit

Project 3 South Carolina: \$300 million greenfield

PROJECT DURATION: 2022 (ongoing)

OUR EXECUTING OFFICES: Appleton, WI, Greenville, SC and
Mumbai, India (GID)

Project Description / Services Provided

Cirba Solutions collects, transports and processes manufacturing scrap and end-of-life batteries, producing black mass from recycled lithium-ion batteries. It plans to invest more than \$1 billion over the next five years to expand the infrastructure for battery recycling. Cirba and Jacobs have formed a strategic alliance to execute this investment program for three planned projects across the US.

Project 1 in Lancaster, OH: We are designing and delivering a \$200 million expansion for end-of-life battery processing facilities on Cirba's Lancaster, OH campus. Our subject matter experts are assisting their engineers and third-party consultants with developing first-of-its-kind equipment and process. We provide staffing to accommodate any client need, including standards development, equipment code expertise, process code expertise, laboratory expertise, and construction management. We are overlapping building development and process/equipment design to maintain the client's number one priority – speed-to-market. Additionally, we assisted Cirba in generating and maintaining procurement procedures that tied to federal grant funding. This heavily phased project overlapped multiple design, bid, construction, and permit packages to drive schedule and maintain completion milestones.

Project 2 in Eloy, AZ: We have started the design of a lithium-ion battery recycling facility in Eloy, AZ. Cirba will move into an already-existing building in Eloy, following a construction project that will expand it by 75,000 feet. The company expects the facility will be in operation by 2023, bringing 110 permanent jobs to the area.

Project 3 in South Carolina: Cirba is investing more than \$300 million in a lithium-ion EV battery recycling plant in South Carolina that will produce materials to power over 500,000 batteries every year. The plant will focus on processing end-of-life hybrid and electric-vehicle batteries, gigafactory scrap and end-of-life consumer batteries to extract critical materials like nickel, cobalt and lithium. Operations are expected to begin in late 2024 at the 200-acre plant located near Columbia, SC.

Thank you!

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